UNIVERSIDAD DE LAS FUERZAS ARMADAS ESPE

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career: Information Technology Engineering

DATE: 05/07/2026

UML DIAGRAMS

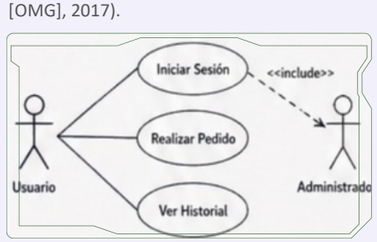
USED IN SOFTWARE ANALYSIS AND DESIGN

The Unified Modeling Language (UML) is a visual modeling language used to represent, analyze, design, and document software systems before their implementation. It facilitates communication among analysts, designers, developers, and clients through standardized diagrams (Object Management Group [OMG], 2017).

Use Case Diagram

Description:

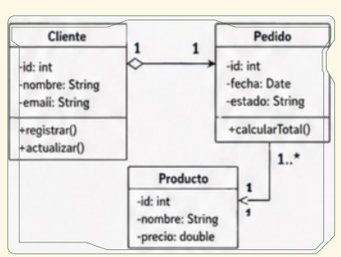
A Use Case Diagram shows the functionality of a system from the user's perspective. It represents the interactions between actors and use cases (Object Management Group [OMG], 2017).



Class Diagram

Description:

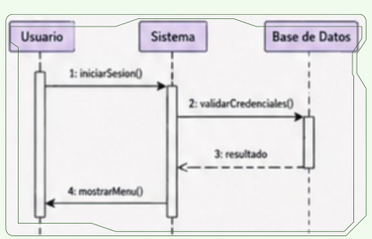
RRepresents the static structure of a system by showing classes, their attributes, methods, and relationships (Fowler, 2004).



Sequence Diagram

Description:

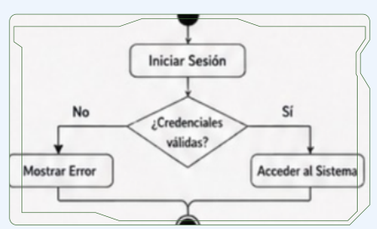
Illustrates the chronological order of messages exchanged between objects to accomplish a specific task (Object Management Group [OMG], 2017).



Activity Diagram

Description:

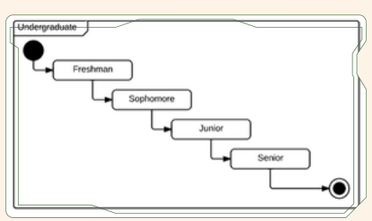
DDescribes the workflow of activities and decisions within a process or business operation (Dennis, Wixom, & Tegarden, 2020).



State Machine Diagram

Description:

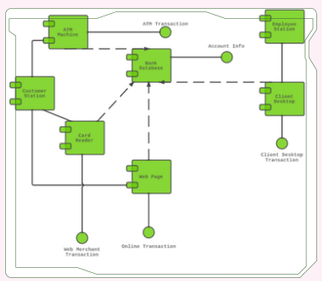
A Use Case Diagram shows the functionality of a system from the user's perspective. It represents the interactions between actors and use cases (Object Management Group [OMG], 2017).



Component Diagram

Description:

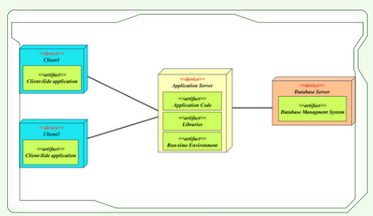
Displays the organization of software components and the dependencies between them (Booch, Rumbaugh, & Jacobson, 2005).



Deployment Diagram

Description:

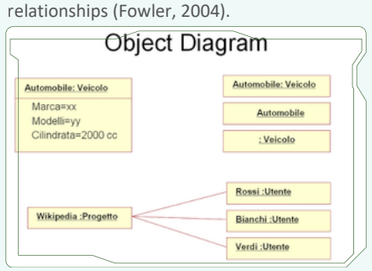
Illustrates how software artifacts are deployed on hardware nodes and the physical architecture of the system (OMG, 2017).



Object Diagram

Description:

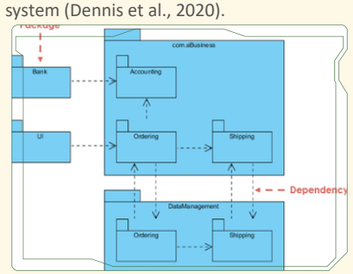
Represents instances of classes at a specific moment, showing objects and their relationships (Fowler, 2004).



Package Diagram

Description:

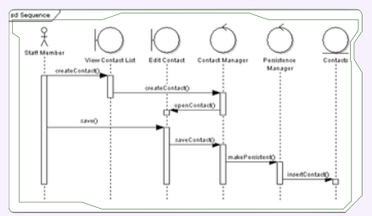
Organizes model elements into packages and shows the dependencies among them, improving the modular structure of the system (Dennis et al., 2020).



Communication Diagram

Description:

Shows how objects collaborate by exchanging messages to perform a specific function, emphasizing object relationships rather than message timing (Booch, Rumbaugh, & Jacobson, 2005).



LITERATURE

1. Graby Booch, G., James Rumbaugh, J., & Ivar Jacobson, I. (2005). The Unified Modeling Language User Guide (2nd ed.). Addison-Wesley.
2. Martin Fowler. (2004). UML Distilled: A Brief Guide to the Standard Object Modeling Language (3rd ed.). Addison-Wesley.
3. Alan Dennis, Barbara Haley Wixom, B. H., & David Tegarden, D. (2020). Systems Analysis and Design: An Object-Oriented Approach with UML (6th ed.). Wiley.
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